Identify Info

FTL:

```
wesan@wesan-VMware-Virtual-Platform:~/nvmevirt$ sudo nvme id-ns /dev/nvme0n1 -H
NVME Identify Namespace 1:
nsze    : 0xff800
ncap    : 0xff800
nuse    : 0xff800
nsfeat  : 0
  [4:4] : 0     NPWG, NPWA, NPDG, NPDA, and NOWS are Not Supported
  [3:3] : 0     NGUID and EUI64 fields if non-zero, Reused
  [2:2] : 0     Deallocated or Unwritten Logical Block error Not Supported
  [1:1] : 0     Namespace uses AWUN, AWUPF, and ACWU
  [0:0] : 0     Thin Provisioning Not Supported

nlbaf   : 6
flbas   : 0
  [6:5] : 0     Most significant 2 bits of Current LBA Format Selected
  [4:4] : 0     Metadata Transferred in Separate Contiguous Buffer
  [3:0] : 0     Least significant 4 bits of Current LBA Format Selected

mc      : 0
  [1:1] : 0     Metadata Pointer Not Supported
  [0:0] : 0     Metadata as Part of Extended Data LBA Not Supported

dpc     : 0
  [4:4] : 0     Protection Information Transferred as Last Bytes of Metadata Not Supported
  [3:3] : 0     Protection Information Transferred as First Bytes of Metadata Not Supported
  [2:2] : 0     Protection Information Type 3 Not Supported
  [1:1] : 0     Protection Information Type 2 Not Supported
  [0:0] : 0     Protection Information Type 1 Not Supported

dps     : 0
  [3:3] : 0     Protection Information is Transferred as Last Bytes of Metadata
  [2:0] : 0     Protection Information Disabled

nmic    : 0
  [0:0] : 0     Namespace Multipath Not Capable
```

```
rescap  : 0
  [7:7] : 0     Ignore Existing Key - Used as defined in revision 1.2.1 or earlier
  [6:6] : 0     Exclusive Access - All Registrants Not Supported
  [5:5] : 0     Write Exclusive - All Registrants Not Supported
  [4:4] : 0     Exclusive Access - Registrants Only Not Supported
  [3:3] : 0     Write Exclusive - Registrants Only Not Supported
  [2:2] : 0     Exclusive Access Not Supported
  [1:1] : 0     Write Exclusive Not Supported
  [0:0] : 0     Persist Through Power Loss Not Supported

fpi     : 0
  [7:7] : 0     Format Progress Indicator Not Supported

dlfeat  : 0
  [4:4] : 0     Guard Field of Deallocated Logical Blocks is set to 0xFFFF
  [3:3] : 0     Deallocate Bit in the Write Zeroes Command is Not Supported
  [2:0] : 0     Bytes Read From a Deallocated Logical Block and its Metadata are Not Reported

nawun   : 0
nawupf  : 0
nacwu   : 0
nabsn   : 0
nabo    : 0
nabspf  : 0
noiob   : 0
nvmcap  : 0
mssrl   : 0
mcl     : 0
msrc    : 0
nulbaf  : 0
anagrpid: 0
nsattr  : 0
nvmsetid: 0
endgid  : 0
nguid   : 00000000000000000000000000000000
eui64   : 0000000000000000
```

```
LBA Format  0 : Metadata Size: 0   bytes - Data Size: 512 bytes - Relative Performance: 0x2 Good (in use)
LBA Format  1 : Metadata Size: 8   bytes - Data Size: 512 bytes - Relative Performance: 0x2 Good
LBA Format  2 : Metadata Size: 16  bytes - Data Size: 512 bytes - Relative Performance: 0x2 Good
LBA Format  3 : Metadata Size: 0   bytes - Data Size: 4096 bytes - Relative Performance: 0 Best
LBA Format  4 : Metadata Size: 8   bytes - Data Size: 4096 bytes - Relative Performance: 0 Best
LBA Format  5 : Metadata Size: 64  bytes - Data Size: 4096 bytes - Relative Performance: 0 Best
LBA Format  6 : Metadata Size: 128 bytes - Data Size: 4096 bytes - Relative Performance: 0 Best
```

ZNS:

```
wesan@wesan-VMware-Virtual-Platform:~/nvmevirt$ sudo nvme zns id-ns /dev/nvme0n1 -H
ZNS Command Set Identify Namespace:
zoc     : 0     Zone Operation Characteristics
  [1:1] : 0       Zone Active Excursions: No
  [0:0] : 0       Variable Zone Capacity: No

ozcs    : 0     Optional Zoned Command Support
  [0:0] : 0       Read Across Zone Boundaries: No
  [1:1] : 0       Zone Random Write Area: No
mar     : 1     Active Resources
mor     : 1     Open Resources
rrl     : Not Reported
frl     : Not Reported
rrl1    : Not Reported
rrl2    : Not Reported
rrl3    : Not Reported
frl1    : Not Reported
frl2    : Not Reported
frl3    : Not Reported
numzrwa : 0
zrwafg  : 0
zrwasz  : 0
zrwacap : 0     Zone Random Write Area Capability
  [0:0] : 0       Explicit ZRWA Flush Operations: No
LBA Format Extension  0 : Zone Size: 0x0 LBAs - Zone Descriptor Extension Size: 0 bytes (in use)
LBA Format Extension  1 : Zone Size: 0x0 LBAs - Zone Descriptor Extension Size: 7168 bytes
LBA Format Extension  2 : Zone Size: 0x7ffd2aa7aaa0 LBAs - Zone Descriptor Extension Size: 2880 bytes
LBA Format Extension  3 : Zone Size: 0x10 LBAs - Zone Descriptor Extension Size: 0 bytes
LBA Format Extension  4 : Zone Size: 0x0 LBAs - Zone Descriptor Extension Size: 9216 bytes
LBA Format Extension  5 : Zone Size: 0x7ffd2aa7aae0 LBAs - Zone Descriptor Extension Size: 704 bytes
LBA Format Extension  6 : Zone Size: 0x7ffd2aa7aaf0 LBAs - Zone Descriptor Extension Size: 8192 bytes
```

```
wesan@wesan-VMware-Virtual-Platform:~/nvmevirt$ sudo nvme zns report-zones /dev/nvme0n1 -n
report-zones: option requires an argument -- 'n'
Usage: nvme zns report-zones <device> [OPTIONS]

Retrieve the Report Zones data structure

Options:
  [  --namespace-id=<NUM>, -n <NUM> ]   --- Namespace identifier to use
  [  --start-lba=<IONUM>, -s <IONUM> ]  --- starting LBA of the zone
  [  --descs=<NUM>, -d <NUM> ]          --- number of descriptors to retrieve
                                            (default: all of them)
  [  --state=<NUM>, -S <NUM> ]          --- state of zones to list
  [  --output-format=<FMT>, -o <FMT> ]  --- Output format: normal|json|binary
  [  --verbose, -v ]                    --- show report zones verbosity
  [  --extended, -e ]                   --- set to use the extended report
                                            zones
  [  --partial, -p ]                    --- set to use the partial report
```

Benchmark

FTL:

執行 Random 讀寫 4KB:

```
wesan@wesan-VMware-Virtual-Platform:~/nvmevirt$ sudo fio --name=ftl-randrw \
  --filename=/dev/nvme0n1 --direct=1 --ioengine=libaio \
  --rw=randrw --rwmixread=50 --bs=4k \
  --iodepth=32 --numjobs=4 --time_based=1 --runtime=30 --group_reporting
ftl-randrw: (g=0): rw=randrw, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=libaio, iodepth=32
...
fio-3.36
Starting 4 processes
Jobs: 4 (f=4): [m(4)][100.0%][r=111MiB/s,w=111MiB/s][r=28.5k,w=28.4k IOPS][eta 00m:00s]
ftl-randrw: (groupid=0, jobs=4): err= 0: pid=3139: Tue Oct 21 16:50:48 2025
  read: IOPS=18.2k, BW=71.1MiB/s (74.5MB/s)(2133MiB/30004msec)
    slat (nsec): min=1830, max=407062, avg=2926.42, stdev=3540.00
    clat (usec): min=91, max=25066, avg=3506.55, stdev=1503.49
     lat (usec): min=125, max=25068, avg=3509.47, stdev=1503.47
    clat percentiles (usec):
     |  1.00th=[ 1500],  5.00th=[ 1975], 10.00th=[ 2212], 20.00th=[ 2540],
     | 30.00th=[ 2802], 40.00th=[ 3032], 50.00th=[ 3228], 60.00th=[ 3458],
     | 70.00th=[ 3687], 80.00th=[ 4015], 90.00th=[ 4752], 95.00th=[ 6718],
     | 99.00th=[ 9372], 99.50th=[10290], 99.90th=[11469], 99.95th=[12911],
     | 99.99th=[24773]
   bw (  KiB/s): min=25032, max=113272, per=99.23%, avg=72233.22, stdev=5226.72, samples=236
   iops        : min= 6258, max=28318, avg=18057.49, stdev=1306.69, samples=236
  write: IOPS=18.2k, BW=71.2MiB/s (74.7MB/s)(2137MiB/30004msec); 0 zone resets
    slat (nsec): min=1860, max=451563, avg=3017.48, stdev=3689.99
    clat (usec): min=90, max=25066, avg=3513.48, stdev=1522.71
     lat (usec): min=124, max=25068, avg=3516.50, stdev=1522.68
    clat percentiles (usec):
     |  1.00th=[ 1500],  5.00th=[ 1975], 10.00th=[ 2212], 20.00th=[ 2540],
     | 30.00th=[ 2802], 40.00th=[ 3032], 50.00th=[ 3228], 60.00th=[ 3458],
     | 70.00th=[ 3687], 80.00th=[ 4047], 90.00th=[ 4752], 95.00th=[ 6783],
     | 99.00th=[ 9503], 99.50th=[10290], 99.90th=[11600], 99.95th=[16909],
     | 99.99th=[24773]
```

```
   bw (  KiB/s): min=26168, max=113144, per=99.24%, avg=72372.98, stdev=5243.88, samples=236
   iops        : min= 6542, max=28286, avg=18092.46, stdev=1310.97, samples=236
  lat (usec)   : 100=0.01%, 250=0.41%, 500=0.04%, 750=0.01%, 1000=0.02%
  lat (msec)   : 2=4.86%, 4=73.96%, 10=20.05%, 20=0.62%, 50=0.04%
  cpu          : usr=1.32%, sys=3.56%, ctx=34071, majf=0, minf=57
  IO depths    : 1=0.1%, 2=0.1%, 4=0.1%, 8=0.1%, 16=0.1%, 32=100.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.1%, 64=0.0%, >=64=0.0%
     issued rwts: total=546045,547005,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=32

Run status group 0 (all jobs):
   READ: bw=71.1MiB/s (74.5MB/s), 71.1MiB/s-71.1MiB/s (74.5MB/s-74.5MB/s), io=2133MiB (2237MB), run=30004-30004msec
  WRITE: bw=71.2MiB/s (74.7MB/s), 71.2MiB/s-71.2MiB/s (74.7MB/s-74.7MB/s), io=2137MiB (2241MB), run=30004-30004msec

Disk stats (read/write):
  nvme0n1: ios=542784/543839, sectors=4343952/4350712, merge=0/0, ticks=1815946/1823251, in_queue=3639198, util=99.73%
```

ZNS:

執行 Sequential Write：

fio --zonemode=zbd --rw=write 模擬合法順序寫入模式；

```
wesan@wesan-VMware-Virtual-Platform:~/nvmevirt$ sudo fio --name=zns-seq-write \
  --filename=/dev/nvme0n1 --zonemode=zbd --direct=1 \
  --ioengine=psync --rw=write --bs=256k \
  --iodepth=1 --numjobs=4 --time_based=1 --runtime=30 \
  --group_reporting
zns-seq-write: (g=0): rw=write, bs=(R) 256KiB-256KiB, (W) 256KiB-256KiB, (T) 256KiB-256KiB, ioengine=psync, iodepth=1
...
fio-3.36
Starting 4 processes
Jobs: 4 (f=4): [W(4)][100.0%][w=377MiB/s][w=1508 IOPS][eta 00m:00s]
zns-seq-write: (groupid=0, jobs=4): err= 0: pid=4609: Tue Oct 21 15:03:58 2025
  write: IOPS=1481, BW=370MiB/s (388MB/s)(10.9GiB/30003msec); 5 zone resets
    clat (usec): min=403, max=4051, avg=638.87, stdev=136.72
     lat (usec): min=412, max=4062, avg=650.20, stdev=137.01
    clat percentiles (usec):
     |  1.00th=[  502],  5.00th=[  537], 10.00th=[  553], 20.00th=[  578],
     | 30.00th=[  594], 40.00th=[  603], 50.00th=[  619], 60.00th=[  635],
     | 70.00th=[  652], 80.00th=[  685], 90.00th=[  725], 95.00th=[  766],
     | 99.00th=[  938], 99.50th=[ 1418], 99.90th=[ 2737], 99.95th=[ 3097],
     | 99.99th=[ 3916]
   bw (  KiB/s): min=58688, max=1551319, per=100.00%, avg=919543.74, stdev=130204.61, samples=97
   iops        : min=  227, max= 6057, avg=3590.37, stdev=508.69, samples=97
  lat (usec)   : 500=1.00%, 750=92.49%, 1000=5.67%
  lat (msec)   : 2=0.64%, 4=0.19%, 10=0.01%
  cpu          : usr=7.17%, sys=14.62%, ctx=88752, majf=0, minf=63
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,44464,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
```

```
Run status group 0 (all jobs):
  WRITE: bw=370MiB/s (388MB/s), 370MiB/s-370MiB/s (388MB/s-388MB/s), io=10.9GiB (11.7GB), run=30003-30003msec

Disk stats (read/write):
  nvme0n1: ios=98/44258, sectors=4192/22657536, merge=0/0, ticks=31/14713, in_queue=14745, util=52.76%
```

執行 Random Read：

fio --zonemode=zbd --rw=randread 測試讀取效能。

```
wesan@wesan-VMware-Virtual-Platform:~/nvmevirt$ sudo fio --name=zns-rand-read \
  --filename=/dev/nvme0n1 --zonemode=zbd --direct=1 \
  --ioengine=libaio --rw=randread --bs=4k \
  --iodepth=32 --numjobs=4 --time_based=1 --runtime=30 \
  --group_reporting
zns-rand-read: (g=0): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=libaio, iodepth=32
...
fio-3.36
Starting 4 processes
Jobs: 4 (f=4): [r(4)][100.0%][r=24.7MiB/s][r=6326 IOPS][eta 00m:00s]
zns-rand-read: (groupid=0, jobs=4): err= 0: pid=4638: Tue Oct 21 15:10:58 2025
  read: IOPS=6507, BW=25.4MiB/s (26.7MB/s)(763MiB/30004msec)
    slat (usec): min=32, max=8780, avg=79.96, stdev=74.13
    clat (usec): min=319, max=16064, avg=4797.80, stdev=651.39
     lat (usec): min=386, max=16103, avg=4877.76, stdev=653.06
    clat percentiles (usec):
     |  1.00th=[ 3621],  5.00th=[ 4113], 10.00th=[ 4178], 20.00th=[ 4293],
     | 30.00th=[ 4359], 40.00th=[ 4424], 50.00th=[ 4555], 60.00th=[ 4883],
     | 70.00th=[ 5276], 80.00th=[ 5473], 90.00th=[ 5604], 95.00th=[ 5735],
     | 99.00th=[ 6390], 99.50th=[ 6849], 99.90th=[ 7635], 99.95th=[10552],
     | 99.99th=[14877]
   bw (  KiB/s): min=24479, max=28127, per=100.00%, avg=26084.75, stdev=824.23, samples=59
   iops        : min= 6119, max= 7031, avg=6520.83, stdev=206.05, samples=59
  lat (usec)   : 500=0.01%, 750=0.01%, 1000=0.01%
  lat (msec)   : 2=0.01%, 4=2.37%, 10=97.57%, 20=0.05%
  cpu          : usr=17.40%, sys=7.52%, ctx=143, majf=0, minf=91
  IO depths    : 1=0.1%, 2=0.1%, 4=0.1%, 8=0.1%, 16=0.1%, 32=100.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.1%, 64=0.0%, >=64=0.0%
     issued rwts: total=195246,0,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=32
```

```
Run status group 0 (all jobs):
   READ: bw=25.4MiB/s (26.7MB/s), 25.4MiB/s-25.4MiB/s (26.7MB/s-26.7MB/s), io=763MiB (800MB), run=30004-30004msec

Disk stats (read/write):
  nvme0n1: ios=194455/0, sectors=1555640/0, merge=0/0, ticks=43784/0, in_queue=43784, util=82.68%
```

Explore Implementation

API / Command differences

傳統 FTL：基本指令（read, write, flush）。做 Random Read/Write。

ZNS：除了 read, write, flush 外，多了 zone_append、zone_mgmt_send/recv（Open/Close/Finish/Reset/Report）。Host 透過 ZNS 指令直接操控 zone life cycle 與 sequential write。

Data placement policies

FTL：裝置端決定 PPA，維護 L2P（maptbl/rmap），支援覆寫；以 wordline 聚合、多通道提升平行度；產生 GC / write amplification。

ZNS：Host/File system 決定資料放在哪個 zone 並 sequential 放置。

無 page-level L2P、無傳統 GC。

Zone management vs. hidden FTL

Hidden FTL：GC、victim line 選擇、有效頁搬移、erase 全在裝置內部「黑箱」處理，主機無感但延遲抖動大、尾延遲易升高。

Zone Management：Host 端控制資料寫進 NAND 的 FTL，SSD controller 減少負擔。且 zone 可分別對應不同 application 應用，不同 zone 之間的數據是獨立的，也就是說用戶可以依照實際需求來設定。以 Sequential Write 策略達到穩定延遲跟降低放大量。